

Nikolai Kulikov

Senior Mechanical Design / R&D Engineer | Prototyping | Deep Tech Hardware

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Work Authorization: United States (EAD / Green Card Pending). No sponsorship required.

Professional Profile

Senior Mechanical Design / R&D Engineer with 10+ years of experience developing complex mechanical systems from concept through prototyping, test iteration, and production support. Hands-on expertise in CAD/CAE, additive manufacturing, DFM/DFA, material selection, and root cause analysis. Experienced in cross-functional collaboration and creating engineering documentation for reliability in demanding environments. Recognized as a Hello Tomorrow Deep Tech Pioneer (2024); inventor on multiple U.S. patent filings.

Core Skills

- **Mechanical Design:** concept development, mechanisms, complex assemblies, design reviews, integration constraints
- **CAD/CAE:** SolidWorks, Autodesk Inventor, AutoCAD; FEA for design verification (fundamentals)
- **Prototyping & Validation:** rapid prototyping (FDM/SLA), test planning, fixture/tooling concepts, design iteration loops
- **DFM/DFA:** manufacturability-driven redesign, injection molding fundamentals, CNC basics, tolerance awareness
- **Materials & Processes:** polymers/composites (exposure), material selection, joining concepts, basic process planning
- **RCA / Reliability:** troubleshooting, failure analysis, corrective actions, design improvements
- **Documentation:** engineering drawings, technical reports, requirements/roadmaps, stakeholder-ready technical summaries
- **Standards (exposure):** GD&T (ASME Y14.5)

Professional Experience

Feb 2026 – Present | Mercor (Mercor.io Corporation) | Engineering Expert (Contractor) | Remote

- Contribute to AI-related project work for Mercor clients; scope and responsibilities vary by client project.
- Provide engineering expertise to support technical deliverables, analysis, and documentation as required.
- Collaborate remotely with client stakeholders and Mercor team to execute assigned tasks within project timelines.

Sep 2024 – Aug 2025 | SkyForge Inc. | Mechanical Lead / R&D Engineer (formerly CTO)

- Led early-stage mechanical R&D for aircraft-adjacent ozone generation / injection hardware concepts (mechanical integration, constraints, and feasibility).
- Coordinated engineering direction, technical documentation, and partner/regulatory alignment support.
- Built concept-level designs and supported planning for test setups and validation approach.

Jan 2023 – Apr 2024 | Seattle Foundation Repairs and BunkerUSA | Senior Mechanical Design Engineer

- Designed custom mechanical solutions and assemblies for protective structures and specialized enclosures for harsh / hazardous environments.
- Supported R&D for structural products: concept, detailed design, materials/process selection, and engineering documentation.
- Coordinated stakeholder requirements, installation constraints, and project deliverables; supported documentation/patent needs.

Aug 2020 – Jul 2021 | PJSC "766 PTSM" | Senior Mechanical Design Engineer

- Led mechanical design of autonomous mobile robotic platforms and remote-controlled electromechanical systems for hazardous environments.
- Engineered structural/protective enclosures; supported integration across mechanical components, sensors, and control interfaces.
- Coordinated engineering activities and documentation for cross-functional alignment and manufacturability.

Aug 2014 – Jul 2019 | JSC "FRPC Titan-Barricades" | Senior Mechanical Design Engineer

- Designed heavy-duty mechanical assemblies and subsystems for large industrial machinery operating under high loads.
- Supported full lifecycle from concept through manufacturing and testing; collaborated with production teams on robustness improvements.
- Led root cause analysis / failure investigations for critical subsystems and implemented corrective design actions.

Dec 2019 – Jun 2024 | Kulikov Nikolai LLC | Founder / Lead Engineer (Product Development & Prototyping)

- Designed precision mechanical parts and tooling for prototyping and small-batch production.
- Led iterative test-and-improve cycles to refine durability and performance under real-world conditions.
- Developed process documentation and supported small-batch production setup; managed partner/client coordination.
- Delivered end-to-end product development: requirements → design → prototype → manufacturing support.

Feb 2024 – Present | Kulicage Dynamics LLC (formerly SynthoShield LLC) | Lead R&D Engineer (Mechanical / Product Development)

- Lead mechanical R&D across deep-tech product lines (energy / advanced materials), from requirements to concept design and prototype iteration.
- Drive build-test-improve cycles; coordinate priorities across engineering tasks and external constraints.
- Develop mechanical layouts, assemblies, and prototype-ready designs; support design verification and feasibility assessments.
- Produce technical documentation for experiments, prototypes, and stakeholder communication.

Education

Volgograd State Technical University (Volgograd, Russia) | 2014

Master's degree equivalent (WES)

Major: **Rocket and Spacecraft Launch and Technical Complexes Engineering** (*Space Launch & Ground Systems Engineering*)

Awards & Recognition

- **Hello Tomorrow — Deep Tech Pioneer (2024)**

Patents

- Radioisotope-induced plasma generator with magnetic field control and multi-mode energy extraction — **Provisional patent (USA, 2025)**
- Plasma Power Generator — **Patent pending (USA, 2025)**
- Stratospheric ozone generator and injector — **Provisional patent (USA, 2024)**
- Composite lightweight protection material — **Provisional patent (USA, 2024)**

Additional

- Additive manufacturing: FDM/SLA prototyping; CNC/injection molding fundamentals
- Available portfolio / technical write-ups upon request